EUROPEAN JOURNAL OF PSYCHOTRAUMATOLOGY
2026, VOL. 17, NO. 1, 2690732
https://doi.org/10.1080/20008066.2026.2690732

EUROPEAN JOURNAL OF PSYCHO-TRAUMATOLOGY
THE OFFICIAL JOURNAL OF THE EUROPEAN SOCIETY FOR TRAUMATIC STRESS STUDIES

Taylor & Francis by Informa

BASIC RESEARCH ARTICLE

OPEN ACCESS Check for updates

Latent classes of adverse childhood experiences and their associations with impulsivity and mental health among adults in Saudi Arabia

Saleh A. Alghamdi

Department of Psychiatry, College of Medicine, Imam Mohammad Ibn Saud Islamic University (IMSIU), Prince Mohammed Ibn Salman Ibn Abdulaziz Road, Riyadh, Saudi Arabia

ABSTRACT

Background: Adverse childhood experiences (ACEs) are associated with later mental-health difficulties, but simple ACE counts can obscure which adversities co-occur within individuals. Person-centred modelling can identify groups of people with similar exposure patterns.

Objective: This cross-sectional study identified latent classes of ACEs among adults living in Saudi Arabia and examined their associations with five facets of UPPS-P impulsive behaviour and brief mental-health indicators.

Methods: An online convenience sample of 379 adults aged 18–65 years completed the 10-item ACE screener, the Arabic Short UPPS-P Impulsive Behaviour Scale, and single-item measures of lifetime psychiatric diagnosis and perceived health impact. Respondents reporting no ACEs were retained as a structural no-ACE reference group. Latent class analysis was conducted among ACE-exposed participants. Model selection considered information criteria, entropy, average posterior probabilities, class size, and interpretability. Alternative class specifications were examined in sensitivity analyses. False discovery rate (FDR) correction was applied to pairwise outcome tests.

Results: Of 379 eligible respondents, 133 (35.1%) reported no ACEs. Among 246 ACE-exposed respondents, a two-class solution was selected and yielded limited/mixed ACE ($n = 140$, 36.9% of the full sample) and poly-victimisation ($n = 106$, 28.0%) classes. The poly-victimisation class showed high probabilities of emotional abuse, physical abuse, witnessed domestic violence, emotional neglect, and household mental illness. Compared with the no-ACE group, the poly-victimisation class had higher negative urgency ($d = 0.55$, $q < .001$), lack of perseverance ($d = 0.43$, $q = .007$), psychiatric diagnosis (42.5% vs 10.5%; OR = 6.27, 95% CI [3.19, 12.31]), and perceived health impact (77.4% vs 21.1%; OR = 12.81, 95% CI [6.91, 23.75]). Family-conflict interactions were exploratory and not statistically robust.

Conclusions: A poly-victimisation class was associated with greater emotion-related impulsivity and mental-health burden. The findings support cautious use of pattern-level ACE assessment alongside ACE counts, while recognising the limitations of cross-sectional convenience sampling and brief outcome measures.

Clases latentes de experiencias adversas en la infancia y sus asociaciones con la impulsividad y la salud mental en adultos de Arabia Saudita

Antecedentes: Las experiencias adversas en la infancia (ACEs por sus siglas en inglés) se asocian con problemas de salud mental en etapas posteriores, pero los recuentos simples de ACEs pueden ocultar qué adversidades coocurren en cada individuo. El modelamiento centrado en la persona puede identificar grupos de personas con patrones similares de exposición.

Objetivo: Este estudio de corte transversal identificó clases latentes de ACEs entre adultos residentes en Arabia Saudita y examinó sus asociaciones con cinco facetas de la conducta impulsiva UPPS-P y con indicadores breves de salud mental.

Método: Una muestra por conveniencia en línea de 379 adultos de entre 18 y 65 años completó el cuestionario de cribado de 10 ítems sobre ACEs, la versión árabe abreviada de la Escala UPPS-P de Conducta Impulsiva, y medidas de un solo ítem sobre el diagnóstico psiquiátrico de por vida y el impacto percibido en la salud. Los encuestados que no reportaron ACEs se mantuvieron como grupo de referencia estructural sin ACE, y el análisis de clases latentes se realizó entre los participantes expuestos a ACEs. La selección del modelo consideró criterios de información, entropía, probabilidades posteriores promedio, tamaño de clase e interpretabilidad. Las especificaciones alternativas de clases se evaluaron mediante análisis de sensibilidad. Se aplicó una corrección por tasa de falsos descubrimientos (FDR, por sus siglas en inglés) a las pruebas pareadas de los resultados.

Resultados: De los 379 encuestados, 133 (35,1%) no reportaron ACEs. Entre los 246 encuestados expuestos a ACEs, una solución de dos clases dio como resultado una clase de

ARTICLE HISTORY

Received 26 March 2026
Revised 23 May 2026
Accepted 10 June 2026

KEYWORDS

Adverse childhood experiences; latent class analysis; impulsivity; UPPS-P; mental health; Saudi Arabia; poly-victimisation; cross-sectional study

PALABRAS CLAVES

Experiencias adversas en la infancia; análisis de clases latentes; impulsividad; trauma; UPPS-P; Arabia Saudita; enfoque centrado en la persona; polivictimización; salud mental

HIGHLIGHTS

- A structural no-ACE reference group plus ACE-exposed latent class analysis identified limited/mixed ACE and poly-victimisation classes.
- All five UPPS-P impulsivity facets are reported; family-conflict moderation is treated as exploratory because small cells limited stability.

CONTACT Saleh A. Alghamdi saalghamedi@imamu.edu.sa Prince Mohammed Ibn Salman Ibn Abdulaziz Road, Riyadh 13318, Saudi Arabia
Supplemental data for this article can be accessed online at https://doi.org/10.1080/20008066.2026.2690732.

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

ACEs limitadas/mixtas (n = 140; 36,9%) y una clase de polivictimización (n = 106; 28,0%). La clase de polivictimización mostró altas probabilidades de abuso emocional, abuso físico, exposición a violencia doméstica, negligencia emocional y enfermedad mental en el hogar. En comparación con el grupo sin ACEs, esta clase presentó mayor urgencia negativa (d = 0,55; q < 0,001), falta de perseverancia (d = 0,43; q = 0,007), diagnóstico psiquiátrico (42,5% vs 10,5%; OR = 6,27; IC 95% [3,19, 12,31]), e impacto percibido en la salud (77,4% vs 21,1%; OR = 12,81; IC 95% [6,91, 23,75]). Las interacciones relacionadas con conflicto familiar fueron exploratorias y no estadísticamente robustas.
Conclusión: Una clase de polivictimización se asoció con mayor impulsividad relacionada con las emociones y con una mayor carga en la salud mental. Los hallazgos respaldan el uso cauteloso de la evaluación de los ACEs a nivel de patrones, junto con los recuentos de ACEs, reconociendo al mismo tiempo las limitaciones del muestreo de corte transversal por conveniencia y de las medidas de resultados breves.

1. Background

Adverse childhood experiences (ACEs) include potentially traumatic or stressful events before age 18, such as abuse, neglect, and household dysfunction. ACEs are consistently associated with poorer mental and physical health in adulthood, and meta-analytic evidence indicates that risk tends to increase as the number of adversities accumulates (Felitti et al., 1998; Hughes et al., 2017). In Saudi Arabia, prior ACE research has also reported substantial exposure to childhood adversity and associations with health, risk behaviours, and social determinants (AlHemyari et al., 2022; Almuneef et al., 2014, 2018).

The conventional cumulative ACE score is useful because it provides a simple burden index. It treats very different exposure histories as equivalent when their counts are the same. One person with two ACEs may report parental separation and household mental illness, while another may report emotional abuse and physical abuse. These patterns may differ in timing, chronicity, relational meaning, and links with later functioning. A count score gives the same weight to each item and cannot show which adversities cluster together within the same individuals. These issues are consistent with calls to move beyond cumulative ACE totals and examine dimensions or patterns of adversity (McLaughlin & Sheridan, 2016). Latent class analysis (LCA) addresses this limitation by identifying subgroups that share similar item-response patterns (Collins & Lanza, 2010; Masyn, 2013). Reviews and empirical comparisons suggest that ACE classes often distinguish lower-adversity, household-dysfunction, maltreatment, and poly-victimisation patterns, although the exact number and content of classes depend on the indicators, sample, and cultural context (Lanier et al., 2018; Merians et al., 2019; Wang et al., 2024).

Impulsivity is a relevant adult outcome domain because it cuts across multiple psychiatric and behavioural problems. The UPPS-P model conceptualises impulsive behaviour as five facets: negative urgency, positive urgency, sensation seeking, lack of premeditation, and lack of perseverance (Cyders et al., 2014; Whiteside & Lynam, 2001). Negative and positive urgency are emotion-driven dimensions, referring to rash action under negative or positive affect. Early adversity may be especially relevant to these emotion-related facets because maltreatment and family threat can shape stress reactivity, emotion regulation, and inhibitory control, although cross-sectional survey data cannot test neurobiological mechanisms directly (Teicher & Samson, 2016). A prior study linked ACE classes with impulsivity, but evidence remains limited for how ACE classes relate to specific UPPS-P facets in Middle Eastern samples (Shin et al., 2018).

Saudi Arabia is an important context for person-centred ACE research because most LCA-ACE evidence is derived from Western samples. Existing Saudi studies show that ACEs are not rare, but estimates vary because studies use different instruments, sampling frames, and adversity definitions (AlHemyari et al., 2022; Almuneef et al., 2014, 2018). Social norms surrounding family privacy, stigma, and help-seeking may also influence disclosure of ACEs and psychiatric diagnoses. These factors make it important to interpret both prevalence and class structure cautiously rather than assuming direct comparability with Western samples.

The present study used cross-sectional online survey data from adults living in Saudi Arabia. We aimed to identify ACE classes and examine their associations with all five UPPS-P impulsivity facets, lifetime psychiatric diagnosis, and perceived health impact. We expected that a higher-adversity/poly-victimisation class would show the strongest associations with emotion-driven impulsivity and mental-health indicators. We also examined current family conflict as an exploratory moderator, while recognising that small cells and cross-sectional measurement limit causal interpretation.

2. Methods

Study design and oversight. This cross-sectional, questionnaire-based study examined ACE classes

and their associations with impulsivity and mental-health indicators among adults living in Saudi Arabia. Ethical approval was obtained from the Institutional Review Board of Imam Mohammad Ibn Saud Islamic University (Project No. 717/2024). All participants provided electronic informed consent. Reporting follows STROBE guidance for observational studies (von Elm et al., 2007).

Participants and recruitment. Inclusion criteria were age 18–65 years, current residence in Saudi Arabia, Arabic literacy, electronic informed consent, and questionnaire completion. Recruitment used an open online survey link disseminated between December 2024 and July 2025 through WhatsApp, Telegram, X/Twitter, Facebook, personal networks, university channels, and community groups. No incentives or sociodemographic quotas were used. Because the survey link was open and invitations were redistributed through multiple channels, the number of people who viewed the invitation is unknown and a conventional response rate cannot be calculated. The survey platform recorded 385 submissions. Three respondents did not provide electronic consent and did not proceed to the questionnaire. Among 382 consenting respondents, three were older than the prespecified upper age limit of 65 years and were excluded. The final analytic sample comprised 379 adults aged 18–65 years.

Sample size. The target of approximately 385 respondents was set to support descriptive precision for a large adult population using a 95% confidence level and an approximate 5% margin of error. For LCA, this target was also considered adequate for exploratory binary-indicator class enumeration with a small number of indicators, provided that selected classes were not very small. Model selection therefore prioritised information criteria as well as class size, posterior classification quality, and interpretability.

Sociodemographic variables. Participants reported gender, nationality, broad region of residence in Saudi Arabia, age group, marital status, education, employment status, family monthly income, and living arrangement. Following reviewer guidance and current psychotrauma reporting recommendations, gender categories are described as women and men rather than female and male unless referring to the original survey response options (Langeland & Olff, 2024).

Adverse childhood experiences. The 10 binary ACE items assessed physical neglect, parent loss/separation, household mental illness, household substance abuse, witnessed domestic violence, household incarceration, emotional abuse, physical abuse, emotional neglect, and sexual abuse before age 18. The items correspond to the standard 10-item ACE domains from the original ACE framework (Felitti et al., 1998). Arabic wording was informed by prior Saudi ACE research using Arabic ACE-IQ adaptations (Almuneef et al., 2014; World Health Organization, 2020). The exact 10-item Arabic screener used here should nevertheless be interpreted as an adapted ACE screener rather than a standalone instrument with comprehensive psychometric validation in this sample; this limitation is acknowledged explicitly.

Impulsivity. The Arabic Short UPPS-P Impulsive Behaviour Scale was used to assess five four-item facets: negative urgency, positive urgency, sensation seeking, lack of premeditation, and lack of perseverance (Bteich et al., 2017). Items were scored from 1 to 4 and oriented so that higher scores indicated greater impulsivity. Internal consistency in the analytic sample was acceptable for negative urgency (α = .818), positive urgency (α = .704), sensation seeking (α = .811), and lack of perseverance (α = .816). Lack of premeditation showed lower internal consistency (α = .548); findings for this subscale are interpreted cautiously.

Mental-health indicators. Lifetime psychiatric diagnosis was assessed with the item: 'Have you ever been diagnosed with any psychiatric illness?' Response options included 'No' and several named diagnoses or an 'Other' option. Responses were dichotomised as any psychiatric diagnosis versus none. Perceived health impact was assessed with the item: 'Do you believe that these experiences have affected your health?' Responses were 'Not much', 'To some extent', and 'A lot'. This variable was dichotomised as Not much versus Some/A lot because the latter two categories both indicated perceived impact and because the three-level distribution was sparse for some class comparisons. These single-item indicators are treated as brief exploratory outcomes rather than diagnostic assessments.

Current family conflict. Current family conflict was assessed with the item: 'How often does it happen that people in your family have serious arguments or fights?' Response options were Never, Rarely, Sometimes, Often, and Always. For moderation analyses, responses were collapsed to low (Never/Rarely), medium (Sometimes), and high (Often/Always) to reduce sparse cells. This decision was made before interpreting the class-by-conflict comparisons and is reported because the high-conflict cell in the no-ACE group remained very small.

Data management and missing data. The online survey required responses for the analytic items. Item-level missingness was therefore 0% among consenting respondents who completed the questionnaire. The only exclusions were non-consent and age outside the prespecified eligibility range. Complete-case analyses were used.

Latent class analysis. Because a large proportion of participants reported no ACEs, respondents with ACE

total = 0 were retained as a structural no-ACE reference group. LCA was then conducted among participants endorsing at least one ACE to identify heterogeneity among ACE-exposed respondents rather than allowing the no-ACE subgroup to dominate class formation. Models with one to five ACE-exposed classes were estimated for the 10 binary ACE indicators using maximum likelihood for a multivariate Bernoulli latent class model with multiple random starts. Model selection considered AIC, BIC, sample-size adjusted BIC, entropy, average posterior probabilities, minimum class size, and substantive interpretability (Nylund et al., 2007; Nylund-Gibson & Choi, 2018). Models producing very small classes were not selected as the primary solution because distal comparisons would be unstable. Sensitivity analyses examining alternative class enumeration strategies, including a full-sample three-class and four-class solution, are reported in the Supplementary Material.

Outcome analyses. Participants were assigned to the no-ACE reference group or to their modal ACE-exposed latent class. Classification quality was assessed using entropy and average posterior probabilities. Group differences in continuous UPPS-P facets were examined with one-way analysis of variance and pairwise Welch tests comparing each ACE-exposed class with the no-ACE reference group. Binary outcomes were examined using raw proportions and odds ratios with 95% confidence intervals. False discovery rate correction was applied to pairwise tests; q-values refer to FDR-adjusted *p*-values. Cumulative ACE score and class-grouping models were compared descriptively using adjusted R^2 for continuous outcomes and McFadden pseudo-R^2 for binary outcomes. Family-conflict moderation was examined only as an exploratory analysis because of small cells.

Software. Descriptive statistics, analysis of variance, Welch tests, logistic regression, and FDR-adjustment were conducted in IBM SPSS Statistics version 28.0. Latent class analysis was performed in R version 4.3 using the poLCA package (Linzer & Lewis, 2011). All numeric results were independently verified across the two statistical environments. Figures were prepared in R.

3. Results

Participant flow and sample characteristics. The analytic sample included 379 adults aged 18–65 years. Respondents came from all five broad geographic regions coded in the survey, although the sample was concentrated in the Central Region and should not be treated as nationally representative. Table 1 summarises sample characteristics by person-centred group. The mean ACE count was 2.03 (SD = 2.23); 64.9% endorsed at least one ACE, 35.1% endorsed three or more ACEs, and 24.5% endorsed four or more ACEs.

Latent class enumeration. Of the 379 eligible participants, 133 (35.1%) reported no ACEs and were retained as a structural no-ACE reference group. LCA was then conducted among the 246 ACE-exposed participants. The two-class ACE-exposed model was selected as the primary model. Although the three-class model showed slightly lower BIC, it produced a very small class ($n = 13$), limiting the stability of outcome comparisons. The selected two-class solution had acceptable entropy (.808), high average posterior probabilities (.947 and .944), and no very small classes. Combined with the no-ACE reference group, the final person-centred grouping comprised no ACE ($n = 133$, 35.1%), limited/mixed ACE ($n = 140$, 36.9%), and poly-victimisation ($n = 106$, 28.0%). Table 2 presents ACE-exposed model fit indices, and Table 3 presents item-response probabilities. Alternative class enumeration solutions, including a full-sample three-class and four-class solution, are reported as sensitivity analyses in Supplementary Tables S1 and S2. The substantive ordering of negative urgency, lack of perseverance, and mental-health outcomes was preserved across alternative class specifications, indicating that the primary findings did not depend on the specific class enumeration chosen.

Class interpretation. The limited/mixed ACE class showed moderate probabilities of emotional abuse, parent loss/separation, household mental illness, and emotional neglect, but low probabilities of witnessed domestic violence and physical abuse. The poly-victimisation class showed very high probabilities of emotional abuse, physical abuse, witnessed domestic violence, emotional neglect, and household mental illness. Figure 1 displays the item-response probabilities.

UPPS-P impulsivity facets. Table 4 reports all five UPPS-P facets. The clearest class differences were observed for negative urgency and lack of perseverance. Compared with the no-ACE group, the poly-victimisation class had higher negative urgency (mean = 2.56 vs 2.14; d = 0.55, q < .001) and lack of perseverance (mean = 2.14 vs 1.87; d = 0.43, $q = .007$). The limited/mixed ACE class also showed higher negative urgency than the no-ACE group ($d = 0.32$, $q = .028$). Positive urgency and sensation seeking were modestly higher in the poly-victimisation class, but pairwise comparisons did not survive FDR correction. Lack of premeditation showed no robust class difference. Figure 2 summarises the UPPS-P means and 95% confidence intervals.

Mental-health indicators. Table 5 shows that psychiatric diagnosis and perceived health impact increased across the person-centred groups. Psychiatric diagnosis was reported by 10.5% of the no-ACE group, 26.4% of the limited/mixed ACE class, and

Table 1. Sample characteristics by person-centred adverse childhood experience group.

Characteristic	Total $N = 379$	No ACE $n = 133$	Limited/mixed ACE $n = 140$	Poly-victimisation $n = 106$	p
Gender					.017
Women	222 (58.6)	65 (48.9)	91 (65.0)	66 (62.3)	
Men	157 (41.4)	68 (51.1)	49 (35.0)	40 (37.7)	
Age group					.422
18–25	211 (55.7)	74 (55.6)	80 (57.1)	57 (53.8)	
26–35	80 (21.1)	23 (17.3)	27 (19.3)	30 (28.3)	
36–45	33 (8.7)	13 (9.8)	12 (8.6)	8 (7.5)	
46–55	36 (9.5)	16 (12.0)	15 (10.7)	5 (4.7)	
56–65	19 (5.0)	7 (5.3)	6 (4.3)	6 (5.7)	
Region					.050
Central Region	285 (75.2)	104 (78.2)	103 (73.6)	78 (73.6)	
Eastern Region	30 (7.9)	6 (4.5)	12 (8.6)	12 (11.3)	
Western Region	28 (7.4)	8 (6.0)	11 (7.9)	9 (8.5)	
Southern Region	25 (6.6)	14 (10.5)	6 (4.3)	5 (4.7)	
Northern Region	11 (2.9)	1 (0.8)	8 (5.7)	2 (1.9)	
Education					.370
Middle school or less	1 (0.3)	1 (0.8)	0 (0.0)	0 (0.0)	
High school	79 (20.8)	25 (18.8)	27 (19.3)	27 (25.5)	
Diploma/university degree	265 (69.9)	96 (72.2)	103 (73.6)	66 (62.3)	
Master's/PhD	34 (9.0)	11 (8.3)	10 (7.1)	13 (12.3)	
Employment					.939
Student	188 (49.6)	61 (45.9)	70 (50.0)	57 (53.8)	
Employed	136 (35.9)	50 (37.6)	51 (36.4)	35 (33.0)	
Unemployed	37 (9.8)	15 (11.3)	13 (9.3)	9 (8.5)	
Retired	18 (4.7)	7 (5.3)	6 (4.3)	5 (4.7)	
Family conflict					< .001
Low	185 (48.8)	86 (64.7)	63 (45.0)	36 (34.0)	
Medium	140 (36.9)	43 (32.3)	57 (40.7)	40 (37.7)	
High	54 (14.2)	4 (3.0)	20 (14.3)	30 (28.3)	

Note. Values are *n* (%). *p*-values are from chi-square tests comparing the three groups. Gender categories are reported as women/men following reviewer guidance; the original survey used Female/Male response options. ACE = adverse childhood experience.

Table 2. Latent class enumeration among ACE-exposed participants ($n = 246$).

K	−2LL	AIC	BIC	ssaBIC	Entropy	Modal class sizes	Avg posterior probability	Decision
1	2729.94	2749.94	2784.99	2753.29	–	246	–	Rejected: no heterogeneity
2	2497.74	2539.74	2613.35	2546.78	.808	140 / 106	.947 / .944	Selected: interpretable, adequate size
3	2433.34	2497.34	2609.51	2508.08	.857	129 / 104 / 13	.941 / .927 / .907	Lower BIC but very small class
4	2395.45	2481.45	2632.18	2495.87	.780	81 / 77 / 76 / 12	.943 / .904 / .895 / .840	Very small class; less stable
5	2362.70	2470.70	2659.99	2488.81	.857	77 / 77 / 74 / 10 / 8	.952 / .934 / .915 / .912 / .902	Overfragmented; two tiny classes

Note. Respondents with ACE total = 0 were retained as a structural no-ACE reference group and were not included in this ACE-exposed class enumeration. The two-class model was selected because it produced interpretable classes, adequate entropy, high average posterior probabilities, and no very small classes. The three-class solution had marginally lower BIC but generated a tiny class ($n = 13$). Full-sample and ACE-exposed alternative solutions are reported in Supplementary Tables S1 and S2. −2LL = −2 log likelihood; AIC = Akaike information criterion; BIC = Bayesian information criterion; ssaBIC = sample-size adjusted BIC.

42.5% of the poly-victimisation class. Compared with no ACE, odds ratios were 3.05 (95% CI [1.56, 5.96]) for limited/mixed ACE and 6.27 (95% CI [3.19, 12.31]) for poly-victimisation. Perceived health impact was reported by 21.1%, 46.4%, and 77.4% of the three groups, respectively; corresponding odds ratios were 3.25 (95% CI [1.91, 5.54]) and 12.81 (95% CI [6.91, 23.75]).

Comparison with cumulative ACE score. The revised analysis no longer claims that latent classes outperform ACE counts. In the present data, cumulative ACE score had slightly stronger raw model fit than the three-group LCA classification for several outcomes (Table 6). The class solution should therefore be interpreted as complementary: ACE count captures overall severity, whereas LCA identifies interpretable co-occurrence patterns, especially the poly-victimisation configuration.

Exploratory family-conflict analyses. The family-conflict analyses were reduced in prominence because several cells were small, especially the no-ACE/high-conflict cell ($n = 4$). Class-by-conflict interaction tests were not statistically robust for

Table 3. Estimated ACE item-response probabilities by person-centred group.

ACE item	No ACE	Limited/mixed ACE	Poly-victimisation
Physical neglect	.000	.088	.259
Parent loss/separation	.000	.319	.315
Household mental illness	.000	.265	.505
Household substance abuse	.000	.125	.118
Witnessed domestic violence	.000	.021	.811
Household incarceration	.000	.108	.131
Emotional abuse	.000	.412	.986
Physical abuse	.000	.144	.810
Emotional neglect	.000	.235	.683
Sexual abuse	.000	.110	.202

Note. The no-ACE group is a structural reference group, so item probabilities are zero. Probabilities for the two ACE-exposed classes are model-estimated item-response probabilities from the selected two-class LCA solution.

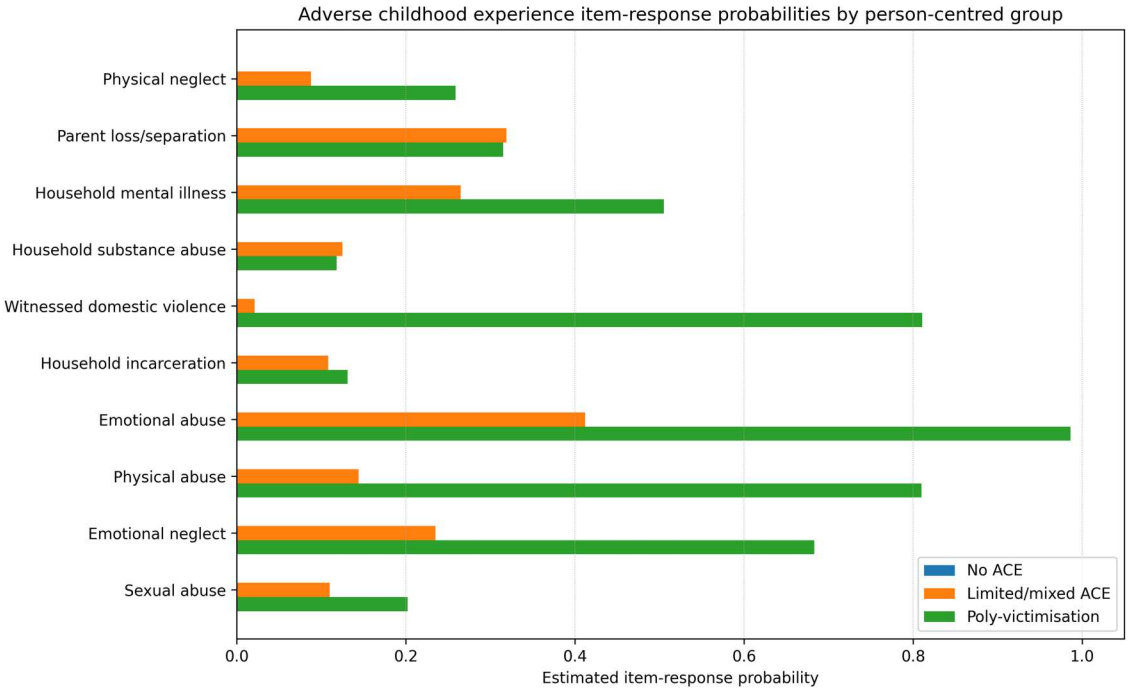


Figure 1. Estimated adverse childhood experience item-response probabilities by person-centred group. The no-ACE group is structural and therefore has zero probability for all ACE indicators. The limited/mixed ACE and poly-victimisation probabilities are from the selected two-class ACE-exposed LCA solution.

negative urgency or lack of perseverance. Table 7 reports the exploratory cell means for transparency, but these analyses should not be interpreted as strong evidence for moderation.

4. Discussion

This study identified person-centred ACE groups among adults living in Saudi Arabia and examined their associations with multidimensional impulsivity and brief mental-health indicators. The revised analysis used a structural no-ACE reference group plus LCA among ACE-exposed respondents. This approach addressed class-enumeration concerns by avoiding an unstable full-sample four-class solution and by preventing the no-ACE group from dominating heterogeneity among exposed participants. The final grouping identified a limited/mixed ACE class and a poly-victimisation class, with the latter showing the highest burden across ACE indicators, emotion-related impulsivity, psychiatric diagnosis, and perceived health impact.

The class structure is consistent with a broad pattern reported in ACE-LCA studies: many samples contain a lower-adversity group and a more severe multi-domain adversity group, while intermediate classes vary by measurement and sample (Lanier et al., 2018; Wang et al., 2024). The current results suggest that in this Saudi online convenience sample, the key distinction among ACE-exposed respondents was between limited/mixed exposure and a high-burden poly-victimisation configuration. This does not mean that household dysfunction or maltreatment-specific classes are absent in the Saudi population. With the present sample size and indicators, attempts to extract more classes produced very small groups that would have been unstable for outcome comparisons (see Supplementary Tables S1–S2).

The prevalence pattern should be interpreted cautiously. The mean ACE count in this sample was 2.03, with 24.5% reporting four or more ACEs. Prior Saudi ACE studies using broader ACE-IQ instruments have also reported substantial adversity exposure, including 32% with four or

Table 4. UPPS-P impulsivity facet scores by person-centred group.

UPPS-P facet	No ACE	Limited/mixed ACE	Poly-victimisation	Overall test	Limited/mixed vs No ACE	Poly-vic. vs No ACE
Negative urgency	2.14 (0.67)	2.36 (0.69)	2.56 (0.84)	$F = 9.65$, $p < .001$	$d = 0.32$, $q = .028$	$d = 0.55$, $q < .001$
Positive urgency	2.60 (0.58)	2.68 (0.62)	2.78 (0.66)	$F = 2.30$, $p = .102$	$d = 0.13$, $q = .385$	$d = 0.28$, $q = .071$
Sensation seeking	2.25 (0.76)	2.26 (0.74)	2.48 (0.82)	$F = 3.22$, $p = .041$	$d = 0.01$, $q = .953$	$d = 0.29$, $q = .071$
Lack of premeditation	1.92 (0.48)	1.88 (0.49)	2.00 (0.52)	$F = 1.81$, $p = .165$	$d = -0.08$, $q = .616$	$d = 0.16$, $q = .370$
Lack of perseverance	1.87 (0.58)	1.88 (0.60)	2.14 (0.68)	$F = 7.14$, $p < .001$	$d = 0.01$, $q = .953$	$d = 0.43$, $q = .007$

Note. Values are mean (SD). Overall tests are one-way ANOVA. Pairwise comparisons use Welch tests against the no-ACE reference group. q-values are FDR-adjusted p-values. Higher scores indicate greater impulsivity. 'ref' / reference group = no ACE.

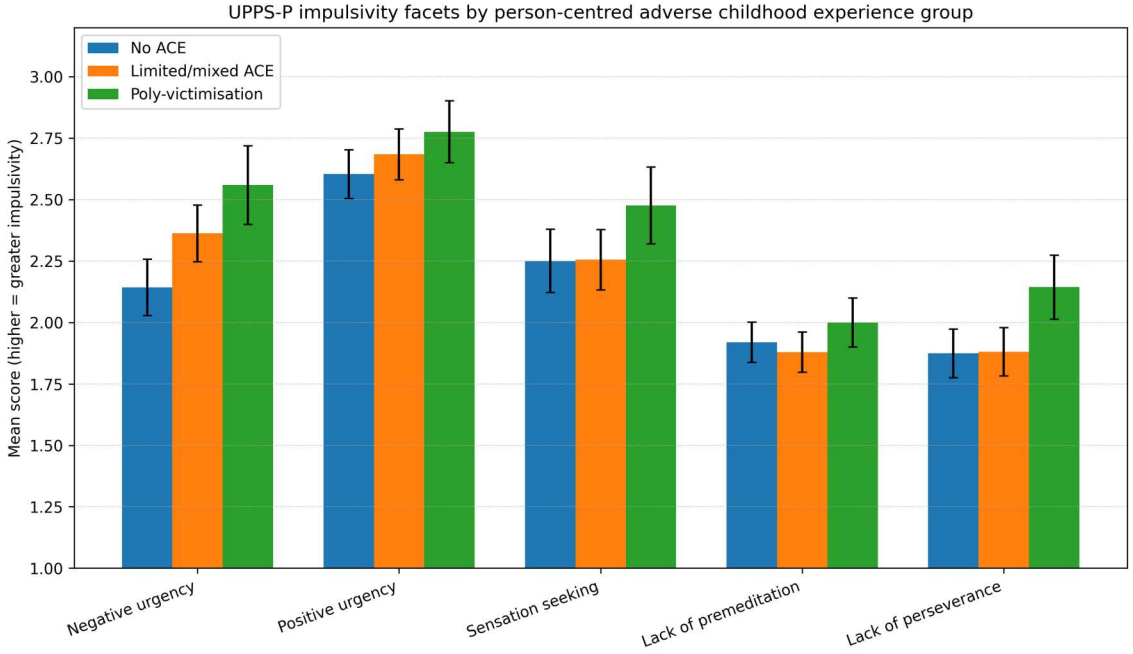


Figure 2. Mean UPPS-P impulsivity facet scores by person-centred adverse childhood experience group. Error bars indicate 95% confidence intervals. Higher values indicate greater impulsivity.

Table 5. Binary mental-health indicators by person-centred group.

Outcome	No ACE	Limited/mixed ACE	Poly-victimisation	Limited/mixed vs No ACE	Poly-vic. vs No ACE
Psychiatric diagnosis	14/133 (10.5)	37/140 (26.4)	45/106 (42.5)	OR = 3.05 [1.56, 5.96] $q = .001$	OR = 6.27 [3.19, 12.31] $q < .001$
Perceived health impact	28/133 (21.1)	65/140 (46.4)	82/106 (77.4)	OR = 3.25 [1.91, 5.54] $q < .001$	OR = 12.81 [6.91, 23.75] $q < .001$

Note. Values are raw *n/N* (%). Odds ratios compare each ACE-exposed class with the structural no-ACE reference group. Psychiatric diagnosis was assessed with a single lifetime diagnosis-history item; perceived health impact was dichotomised as Not much versus Some/A lot. *q*-values are FDR-adjusted *p*-values. CI = 95% confidence interval.

more ACEs in a Riyadh pilot study and high prevalence of multiple ACE categories in a larger national study (Almuneef et al., 2014, 2018). Direct comparisons are difficult because the present study used a 10-item screener and an online convenience sample. The findings nevertheless add person-centred evidence from a Middle Eastern context where ACE-LCA research remains limited.

Table 6. Descriptive comparison of cumulative ACE count and person-centred grouping as predictors of impulsivity and mental-health outcomes.

Outcome	Model	Fit statistic	AIC	BIC
Negative urgency	ACE count	Adj $R^2 = .063$	832.4	840.3
	LCA grouping	Adj $R^2 = .044$	841.2	853.0
Lack of perseverance	ACE count	Adj $R^2 = .031$	711.2	719.1
	LCA grouping	Adj $R^2 = .031$	712.2	724.0
Psychiatric diagnosis	ACE count	McFadden $R^2 = .103$	388.8	396.7
	LCA grouping	McFadden $R^2 = .078$	401.7	413.5
Perceived health impact	ACE count	McFadden $R^2 = .182$	432.0	439.9
	LCA grouping	McFadden $R^2 = .152$	449.7	461.5

Note**.** For continuous UPPS-*P* outcomes, the fit statistic is adjusted R^2 from OLS regression. For binary outcomes, the fit statistic is McFadden pseudo-R^2 from logistic regression. Lower AIC/BIC indicates better fit. This table is included to clarify that the LCA grouping is interpreted as complementary pattern-level information rather than as superior to ACE count for explanatory fit.

The impulsivity results were most robust for negative urgency and lack of perseverance. Negative urgency reflects rash action under negative affect, and its elevation in ACE-exposed classes is consistent with the possibility that childhood adversity is linked with emotion regulation difficulties. Lack of perseverance was elevated mainly in the poly-victimisation class, suggesting that sustained goal-directed effort may be particularly affected among respondents reporting multi-domain adversity. The cross-sectional

Table 7. Exploratory family-conflict cell means.

Person-centred group	Family conflict	*n*	Negative urgency mean (SD)	Lack of perseverance mean (SD)
No ACE	Low	86	2.05 (0.66)	1.92 (0.61)
No ACE	Medium	43	2.34 (0.65)	1.79 (0.51)
No ACE	High	4	2.00 (0.82)	1.69 (0.47)
Limited/mixed ACE	Low	63	2.25 (0.71)	1.80 (0.69)
Limited/mixed ACE	Medium	57	2.42 (0.67)	1.95 (0.53)
Limited/mixed ACE	High	20	2.56 (0.71)	1.95 (0.43)
Poly-victimisation	Low	36	2.38 (0.78)	2.08 (0.60)
Poly-victimisation	Medium	40	2.58 (0.78)	2.09 (0.61)
Poly-victimisation	High	30	2.74 (0.97)	2.28 (0.86)

Note. Family conflict was collapsed to low (Never/Rarely), medium (Sometimes), and high (Often/Always). These analyses are exploratory because some cells were small, particularly no ACE/high conflict ($n = 4$). Class-by-conflict interaction tests were not statistically robust: negative urgency $p > .7$; lack of perseverance $p > .8$.

design cannot establish whether ACEs preceded the impulsivity pattern, whether current mental-health difficulties influenced responses, or whether unmeasured factors contributed to both.

The brief mental-health indicators showed large class differences, especially for the poly-victimisation class. These findings should be read as associations with self-reported diagnosis history and perceived health impact, not as estimates of current psychiatric disorder. The high perceived impact in the poly-victimisation class may reflect subjective recognition of adversity-related health burden, but it could also reflect current distress, recall bias, or differences in willingness to attribute health problems to childhood experiences.

The revised comparison with cumulative ACE score is important. The data did not support the earlier stronger claim that latent classes provided superior explanatory fit. ACE count performed well as an overall severity measure, especially for mental-health indicators. The added value of LCA is more modest and more clinically descriptive: it shows that a sizeable subset of ACE-exposed respondents had a pattern marked by emotional abuse, physical abuse, witnessed domestic violence, emotional neglect, and household mental illness. For trauma-informed screening, this pattern-level information may help clinicians ask more targeted follow-up questions than a count score alone, while still retaining the simplicity of ACE burden.

The family-conflict analysis did not provide consistent evidence of moderation. This does not rule out the relevance of current family environment; rather, the present study was not well powered for this interaction because the no-ACE/high-conflict cell was very small. Future studies should assess current family functioning with validated multi-item instruments and recruit larger samples to test whether current conflict amplifies or buffers associations between ACE classes and adult outcomes (Repetti et al., 2002).

Cultural and social context requires cautious interpretation. In Saudi Arabia and other family-centred societies, ACE disclosure may be shaped by stigma, family privacy, perceived loyalty, and concerns about social judgement. Psychiatric diagnosis may also be underreported if respondents have limited access to diagnosis, avoid mental-health services, or experience stigma. Conversely, online anonymous surveys may allow some respondents to disclose experiences they would not report in interviews. These factors could affect both prevalence estimates and class formation. The overrepresentation of respondents from the Central Region and of young, educated, and student participants further limits generalisability. Underrepresented subgroups might have different ACE patterns, and rare classes may not have emerged because the sample was not nationally representative.

Clinical implications should therefore be specific but cautious. The findings support considering multi-domain ACE exposure, particularly poly-victimisation, when assessing adults who present with emotion-driven impulsivity, difficulty sustaining effort, or mental-health burden. Screening should not rely solely on a total ACE count; however, pattern-level assessment should be conducted sensitively, with attention to privacy, stigma, and referral pathways. The results also suggest that services in Saudi Arabia may benefit from trauma-informed approaches that integrate emotion regulation and impulse-control strategies, especially for individuals reporting multiple forms of childhood maltreatment and household dysfunction.

Limitations. Several limitations qualify the conclusions. First, the cross-sectional design precludes temporal or causal inference. Second, ACEs were retrospectively self-reported and may be affected by recall, reinterpretation, and disclosure bias. Third, the exact 10-item Arabic ACE screener used here was adapted from standard ACE items and prior Arabic ACE-IQ use, but it was not validated as a standalone instrument in this sample. Fourth, psychiatric diagnosis and perceived health impact were single-item indicators, not structured clinical assessments. Fifth, current family conflict was assessed using a single item and the moderation cells were small. Sixth, the online convenience sample was not nationally representative, overrepresented some sociodemographic groups, and did not allow calculation of a conventional response rate. Seventh, the lack of premeditation subscale showed low internal consistency and should be interpreted cautiously. Finally, class solutions depend on the selected indicators and sample; larger representative Saudi and regional studies are needed to test whether the present two-class ACE-exposed solution replicates. Sensitivity analyses across alternative class specifications (Supplementary Tables S1–S2) showed that the substantive ordering of effects was preserved.

Conclusion. A structural no-ACE reference group plus ACE-exposed LCA identified limited/mixed ACE and poly-victimisation classes among adults living in Saudi Arabia. Poly-victimisation was associated with higher negative urgency, lack of perseverance, psychiatric diagnosis, and perceived health impact. The revised findings support using ACE classes as complementary pattern-level information alongside cumulative ACE scores, not as causal or temporally ordered evidence. Future representative studies with validated Arabic measures and stronger mental-health assessment are needed before translating these findings into population-level screening policy.

Acknowledgements

The author thanks all participants who contributed their time and experiences to this study.

Author contributions

CRediT: **Saleh A. Alghamdi:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Deanship of Scientific Research at Imam Mohammad Ibn Saud Islamic University (grant IMSIU-DDRSP2601).

Data availability statement

The de-identified dataset is not publicly posted because it contains sensitive information on childhood adversity, psychiatric history, and substance-related behaviours. De-identified data and analysis syntax are available from the corresponding author on reasonable request, subject to institutional approval and applicable data-protection requirements.

Ethics approval and consent to participate

The Institutional Review Board of Imam Mohammad Ibn Saud Islamic University approved the study (Project No. 717/2024; Session 73; approval date: 1 December 2024). All participants provided electronic informed consent before accessing the questionnaire.

ORCID

Saleh A. Alghamdi http://orcid.org/0000-0003-3264-7397

References

AlHemyari, A. H., Al-Zamil, N. A., Shaikh, A. Y., Al-Eidi, D. A., Al-Dahlan, H. W., & Al-Shamekh, S. S. (2022). Prevalence of adverse childhood experiences and their relationship to mental and physical illnesses in the Eastern Region of Saudi Arabia. *Brain and Behavior*, *12*(7), e2668. https://doi.org/10.1002/brb3.2668

Almuneef, M., ElChoueiry, N., Saleheen, H. N., & Al-Eissa, M. (2018). The impact of adverse childhood experiences on social determinants among Saudi adults. *Journal of Public Health*, *40*(3), e219–e227. https://doi.org/10.1093/pubmed/fdx177

Almuneef, M., Qayad, M., Aleissa, M., & Albuhairan, F. (2014). Adverse childhood experiences, chronic diseases, and risky health behaviors in Saudi Arabian adults: A pilot study. *Child Abuse & Neglect*, *38*(11), 1787–1793. https://doi.org/10.1016/j.chiabu.2014.06.003

Bteich, G., Berbiche, D., & Khazaal, Y. (2017). Validation of the short Arabic UPPS-P impulsive behavior scale. *BMC Psychiatry*, *17*(1), 244. https://doi.org/10.1186/s12888-017-1407-y

Collins, L. M., & Lanza, S. T. (2010). *Latent class and latent transition analysis: With applications in the social, behavioral, and health sciences*. Wiley.

Cyders, M. A., Littlefield, A. K., Coffey, S., & Karyadi, K. A. (2014). Examination of a short English version of the UPPS-P Impulsive Behavior Scale. *Addictive Behaviors*, *39*(9), 1372–1376. https://doi.org/10.1016/j.addbeh.2014.02.013

Felitti, V. J., Anda, R. F., Nordenberg, D., Williamson, D. F., Spitz, A. M., Edwards, V., Koss, M. P., & Marks, J. S. (1998). Relationship of childhood abuse and household dysfunction to many of the leading causes of death in adults: The Adverse Childhood Experiences Study. *American Journal of Preventive Medicine*, *14*(4), 245–258. https://doi.org/10.1016/S0749-3797(98)00017-8

Hughes, K., Bellis, M. A., Hardcastle, K. A., Sethi, D., Butchart, A., Mikton, C., Jones, L., & Dunne, M. P. (2017). The effect of multiple adverse childhood experiences on health: A systematic review and meta-analysis. *The Lancet Public Health*, *2*(8), e356–e366. https://doi.org/10.1016/S2468-2667(17)30118-4

Langeland, W., & Olff, M. (2024). Sex and gender in psychotrauma research. *European Journal of Psychotraumatology*, *15*(1), 2358702. https://doi.org/10.1080/20008066.2024.2358702

Lanier, P., Maguire-Jack, K., Lombardi, B., Frey, J., & Rose, R. A. (2018). Adverse childhood experiences and child health outcomes: Comparing cumulative risk and latent class approaches. *Maternal and Child Health Journal*, *22*(3), 288–297. https://doi.org/10.1007/s10995-017-2365-1

Linzer, D. A., & Lewis, J. B. (2011). poLCA: An R package for polytomous variable latent class analysis. *Journal of Statistical Software*, *42*(10), 1–29. https://doi.org/10.18637/jss.v042.i10

Masyn, K. E. (2013). Latent class analysis and finite mixture modeling. In T. D. Little (Ed.), *The Oxford handbook of quantitative methods in psychology* (Vol. 2, pp. 551–611). Oxford University Press.

McLaughlin, K. A., & Sheridan, M. A. (2016). Beyond adverse childhood experiences: A dimensional approach to developmental psychopathology. *Current Directions in Psychological Science*, *25*(4), 239–245. https://doi.org/10.1177/0963721416655883

Merians, A. N., Baker, M. R., Frazier, P., & Lust, K. (2019). Outcomes related to adverse childhood experiences in college students: Comparing latent class analysis and cumulative risk. *Child Abuse & Neglect*, *87*, 51–64. https://doi.org/10.1016/j.chiabu.2018.07.020

Nylund, K. L., Asparouhov, T., & Muthén, B. O. (2007). Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. *Structural Equation Modeling: A Multidisciplinary Journal*, *14*(4), 535–569. https://doi.org/10.1080/10705510701575396

Nylund-Gibson, K., & Choi, A. Y. (2018). Ten frequently asked questions about latent class analysis. *Translational Issues in Psychological Science*, *4*(4), 440–461. https://doi.org/10.1037/tps0000176

Repetti, R. L., Taylor, S. E., & Seeman, T. E. (2002). Risky families: Family social environments and the mental and physical health of offspring. *Psychological Bulletin*, *128*(2), 330–366. https://doi.org/10.1037/0033-2909.128.2.330

Shin, S. H., McDonald, S. E., & Conley, D. (2018). Profiles of adverse childhood experiences and impulsivity. *Child*

Abuse & Neglect, *85*, 118–126. https://doi.org/10.1016/j.chiabu.2018.07.028

Teicher, M. H., & Samson, J. A. (2016). Annual research review: Enduring neurobiological effects of childhood abuse and neglect. *Journal of Child Psychology and Psychiatry*, *57*(3), 241–266. https://doi.org/10.1111/jcpp.12507

von Elm, E., Altman, D. G., Egger, M., Pocock, S. J., Gøtzsche, P. C., & Vandenbroucke, J. P. (2007). The strengthening the reporting of observational studies in epidemiology (STROBE) statement: Guidelines for reporting observational studies. *PLoS Medicine*, *4*(10), e296. https://doi.org/10.1371/journal.pmed.0040296

Wang, X., Jiang, L., Barry, L., Zhang, X., Vasilenko, S. A., & Heath, R. D. (2024). A scoping review on adverse childhood experiences studies using latent class analysis: Strengths and challenges. *Trauma, Violence, & Abuse*, *25*(2), 1695–1708. https://doi.org/10.1177/15248380231192922

Whiteside, S. P., & Lynam, D. R. (2001). The five factor model and impulsivity: Using a structural model of personality to understand impulsivity. *Personality and Individual Differences*, *30*(4), 669–689. https://doi.org/10.1016/S0191-8869(00)00064-7

World Health Organization. (2020). *Adverse childhood experiences international questionnaire (ACE-IQ)*.